

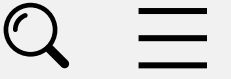
NATIONAL INSTITUTE OF TECHNOLOGY AGARTALA

Department of Production Engineering



MINI PROJECT PRESENTATION

**Voice-Controlled Arduino Writing Machine using a
Custom Web Application**



Voice Scribe

INSTANT VOICE-TO-PAPER PLOTTER

Project Guide : Dr. Debashis Podder
Assistant Professor, Dept. of Production Engineering
National Institute of Technology Agartala

INTRODUCTION



The Context: In academic and legal environments, physical handwriting ("wet ink" on paper) is often mandatory for exams, signatures, and official records.

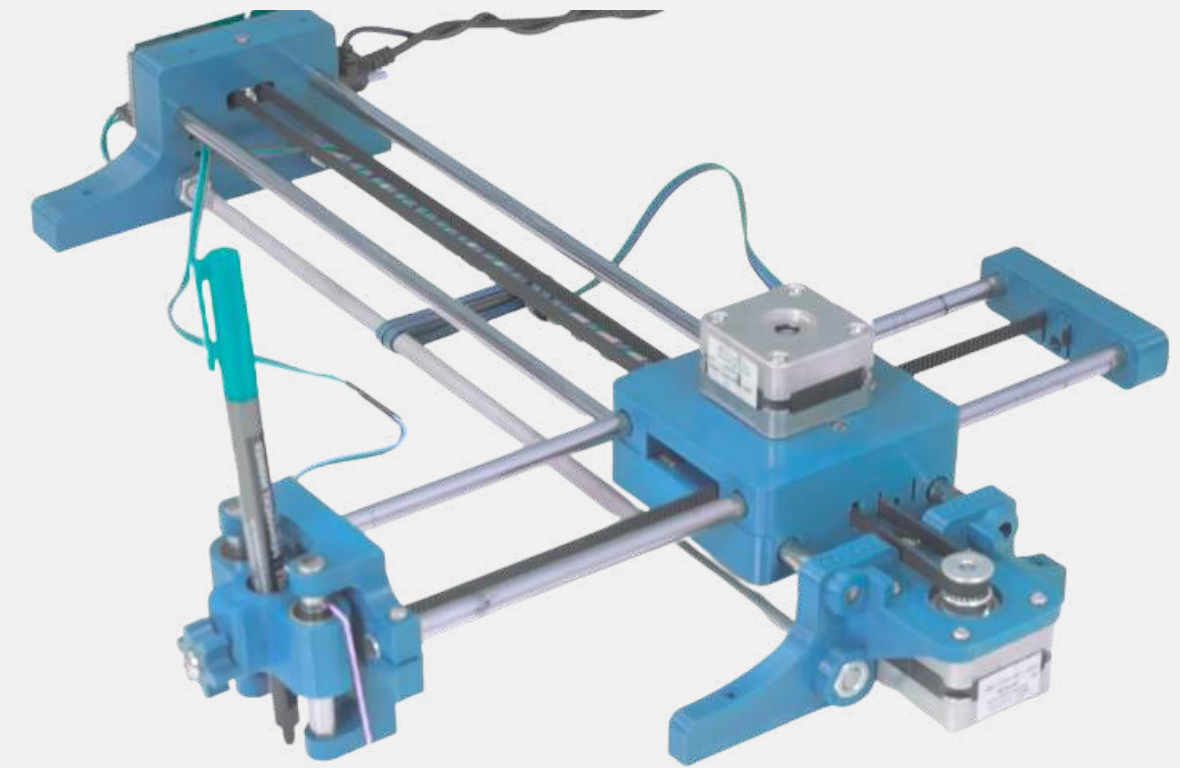
The Problem: According to health statistics, millions of individuals suffer from upper-limb motor impairments (Paralysis, Cerebral Palsy, Amputation) that prevent them from holding a pen.

It eliminates the cost, dependency, and privacy risks of human scribes while overcoming printers' inability to navigate bound notebooks or create legally valid pen indentations.

INTRODUCTION

A 2D CNC writing machine that translates spoken words into physical writing on paper in real-time.

It enables direct, real-time translation of human speech into physical handwritten text on paper, bypassing traditional CAD/CAM workflows.



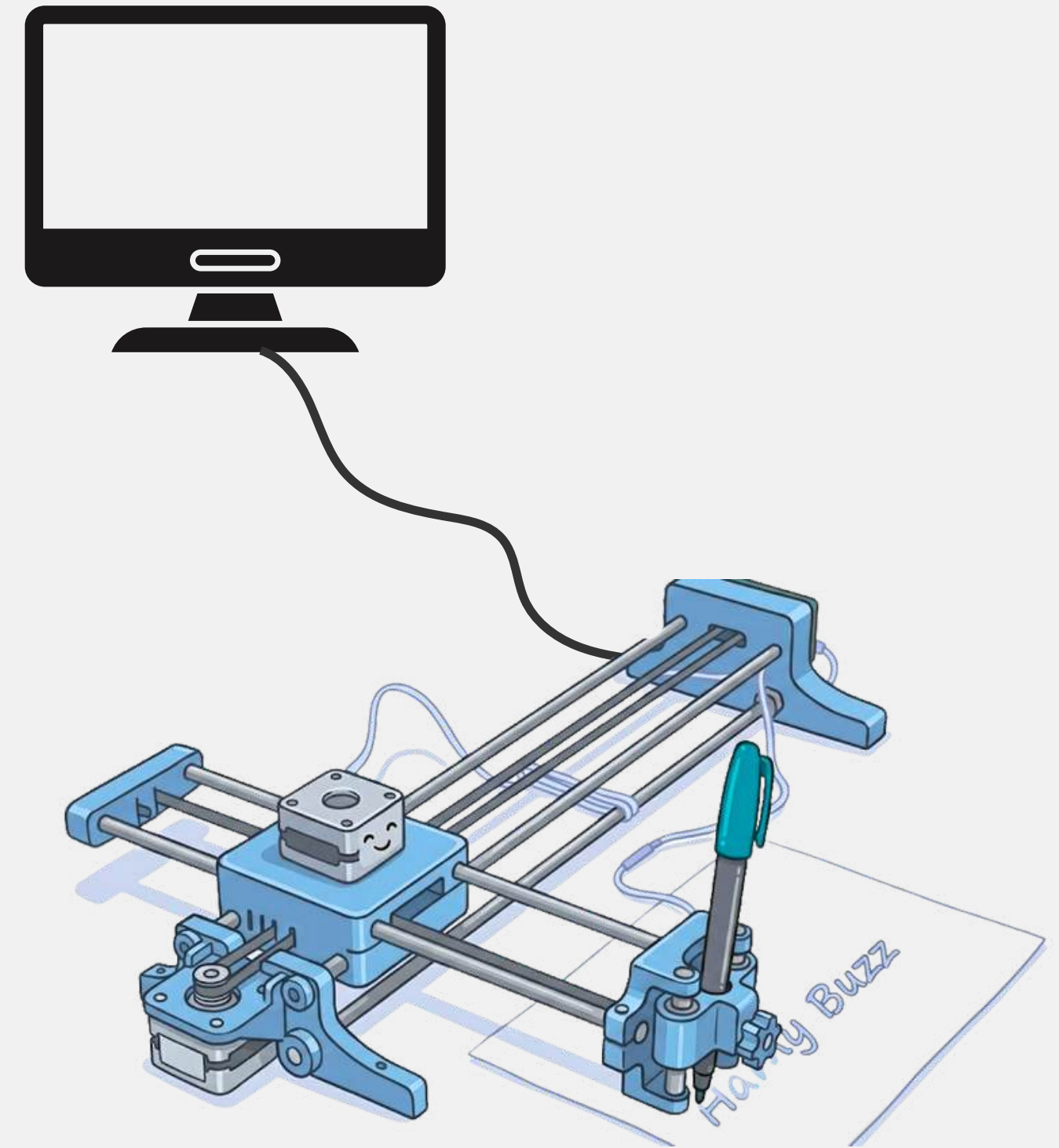
OBJECTIVES



Primary Objective (Social): To develop a hands-free assistive device that allows users to write on standard paper using only voice commands.

Secondary Objective (Technical): To engineer a low-latency Web-Serial interface that eliminates the need for complex CAM software, making the device "plug-and-play.

Economic Objective: To achieve a Low-Cost Design (< ₹4,000 / \$50) using off-the-shelf components and 3D printing,



SYSTEM ARCHITECTURE



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Input Layer

User speaks into a
standard microphone
/headset.



SYSTEM ARCHITECTURE

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Input Layer

- Web Speech API: Converts audio to text strings.
- Vector Engine (JS): Instantly maps text to G-Code coordinates (X, Y, Z).



SYSTEM ARCHITECTURE

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Input Layer

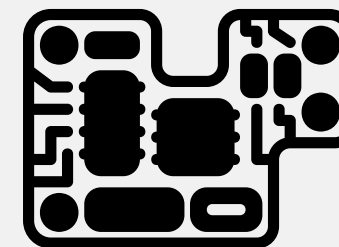
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Web Serial API streams G-code via USB to the microcontroller.



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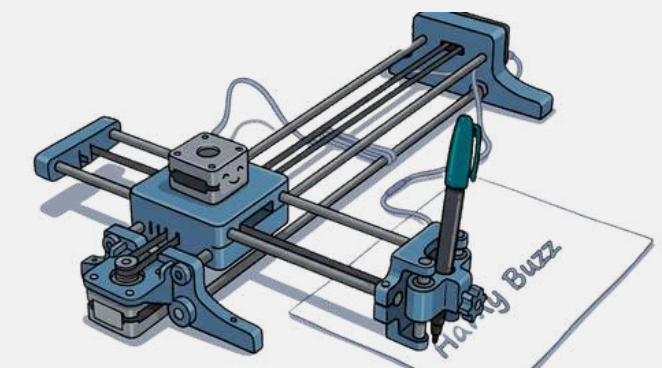
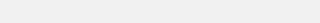
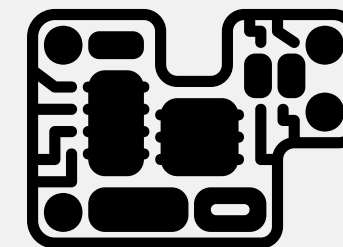
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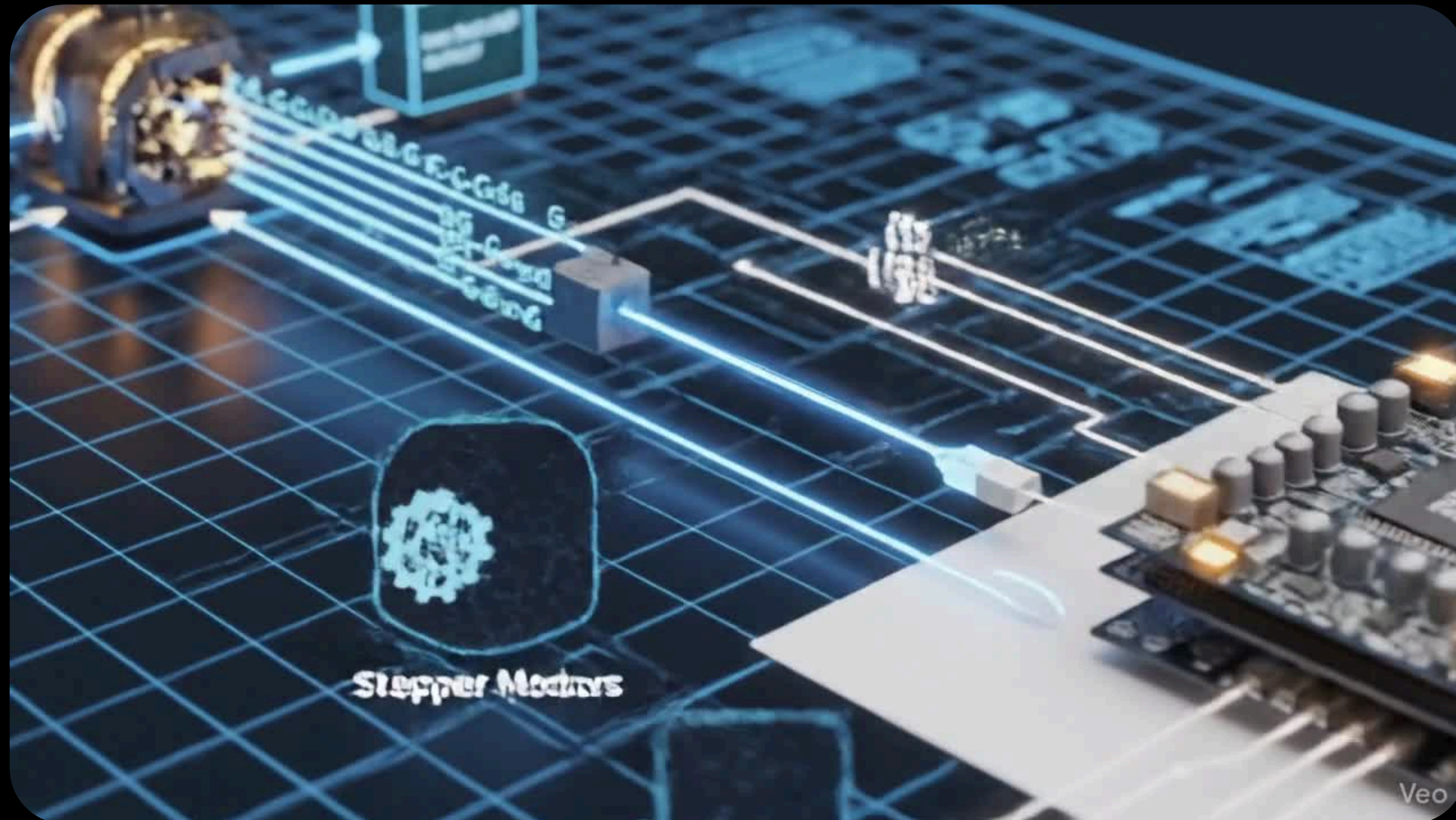
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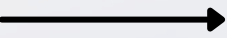
Arduino Uno + CNC Shield drives Steppers (X/Y) and Servo (Z) to manipulate the pen.

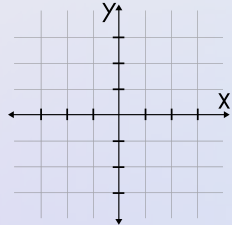


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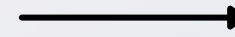


WORKING PRINCIPLE



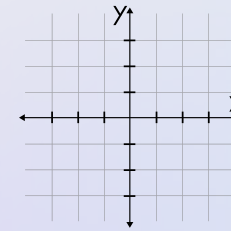
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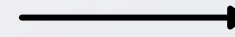
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Vector Translation

Unlike a printer that prints pixels (raster), this machine draws lines (vectors). It treats handwriting as a series of mathematical paths.

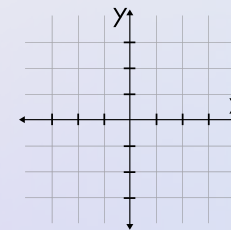
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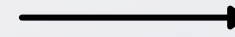
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H-Bot/CoreXY Kinematics

Explaining how two stationary motors pull belts to move the central carriage in 2D space.

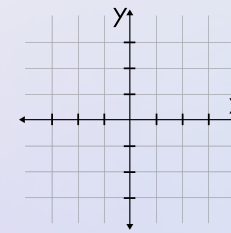
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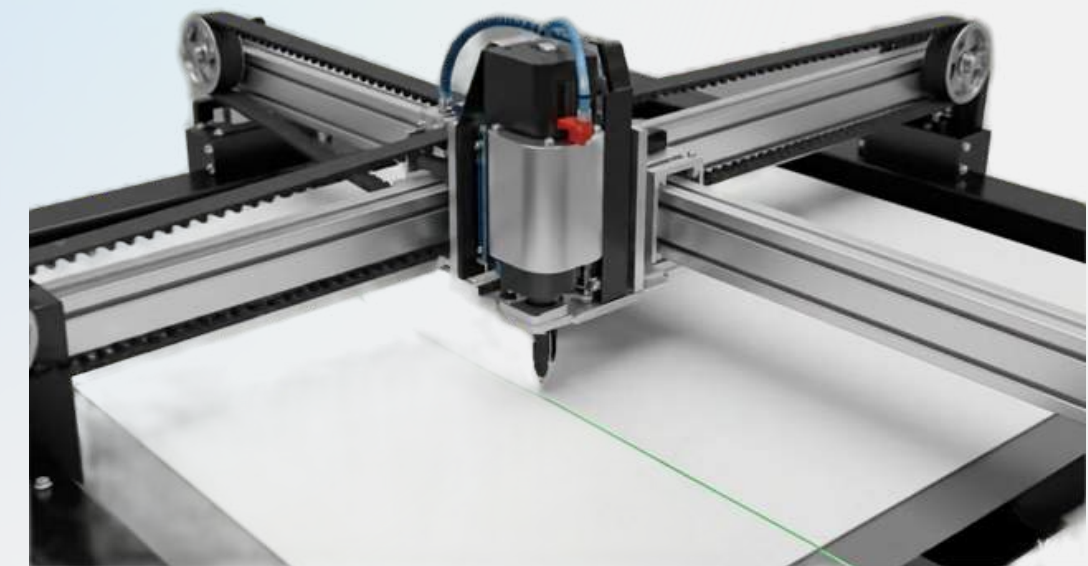
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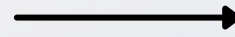
Pen Up/Down State:

Draw State: Z-axis Servo lowers the pen; Steppers move X/Y to create a line.

Travel State: Z-axis Servo lifts the pen; Steppers move X/Y to the next character without marking paper.



ELECTRICAL COMPONENTS



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Arduino Uno R3 (ATmega328P)
(Brain)



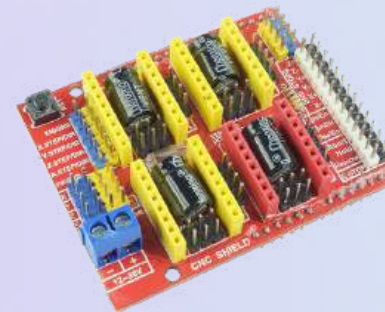
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CNC Shield V3 Expansion Board
(Muscle Driver)



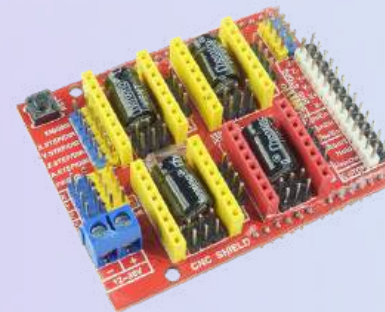
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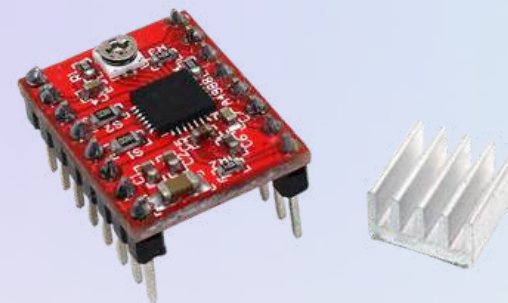
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A4988 Stepper Drivers
(Motor Controllers)



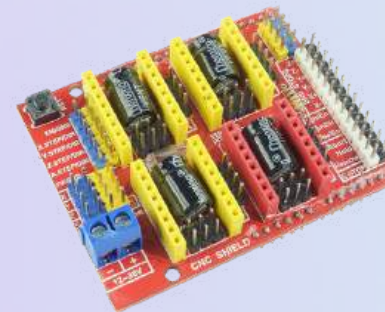
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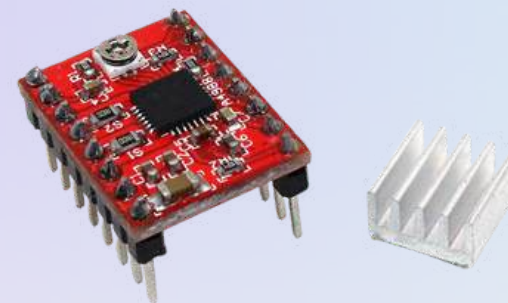
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Actuators

X & Y Axis: NEMA 17 Stepper Motors
(High torque, precise steps).



Z Axis: SG90 Micro Servo (Fast actuation
for pen lifting).



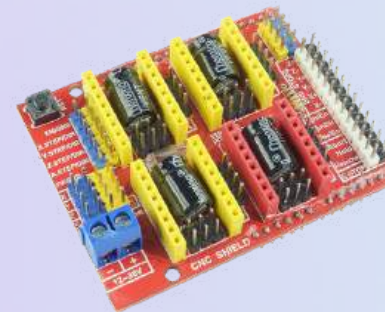
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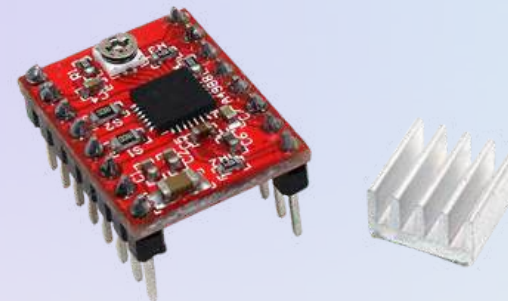
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- **Power: 12V 2A DC Power Supply** (Separated from USB power to prevent voltage sag).



MECHANICAL COMPONENTS

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M8 Threaded Rods + Nuts



MECHANICAL COMPONENTS

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M8 Threaded Rods + Nuts



**GT2 Timing Belts (2mm
pitch)**



MECHANICAL COMPONENTS

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M8 Threaded Rods + Nuts



GT2 Timing Belts (2mm pitch)



GT2 20-Tooth Pulleys & Idlers.



MECHANICAL COMPONENTS

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M8 Threaded Rods + Nuts



GT2 Timing Belts (2mm pitch)



GT2 20-Tooth Pulleys & Idlers.



Linear Motion

8mm Smooth Steel Rods (Chromed for low friction).



LM8UU Linear Ball Bearings.



MECHANICAL COMPONENTS

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M8 Threaded Rods + Nuts



GT2 Timing Belts (2mm pitch)



GT2 20-Tooth Pulleys & Idlers.



Fasteners: M3, M4, and M8 bolts.

Linear Motion

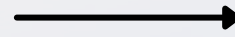
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CUSTOM COMPONENTS



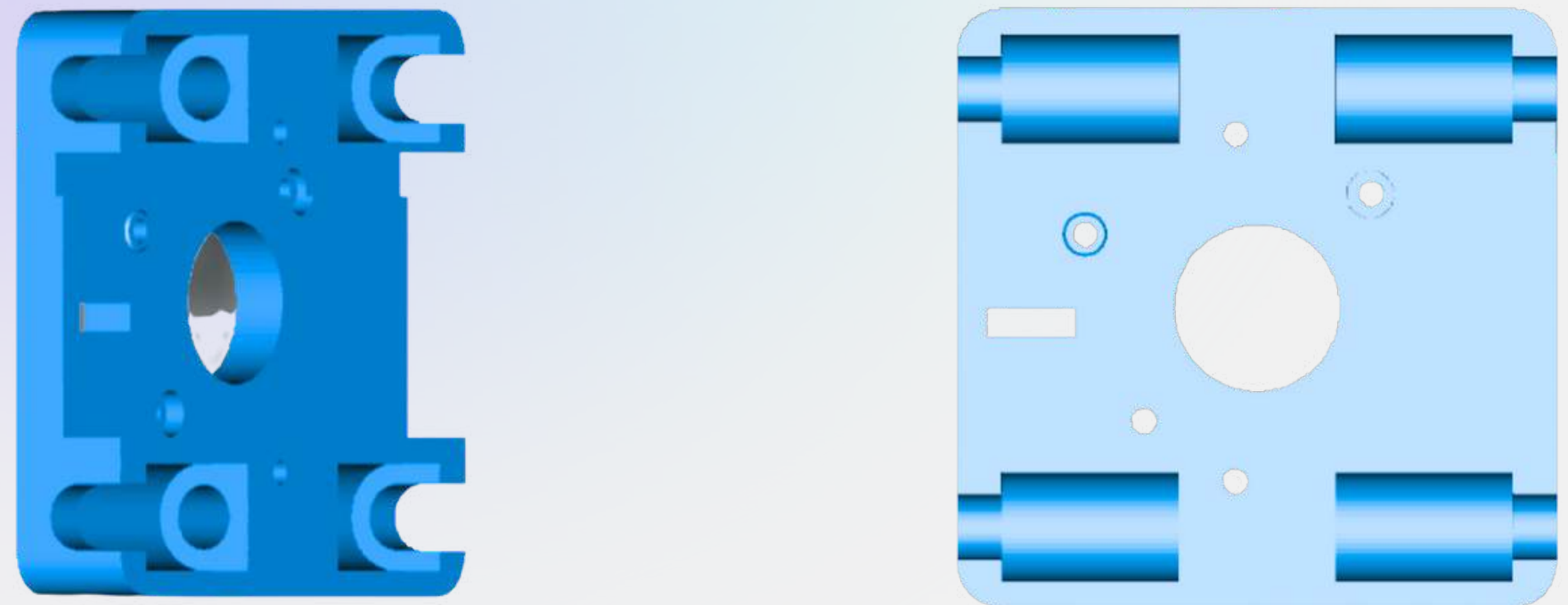
Motor Mounts: Designed to hold NEMA 17 motors securely to the threaded rod frame.

Writting Machine Y axis (Nema17 Stepper Motor Holder Upto 38mm Height) .stl



Writting Machine Y Axis (Arduino Box).stl

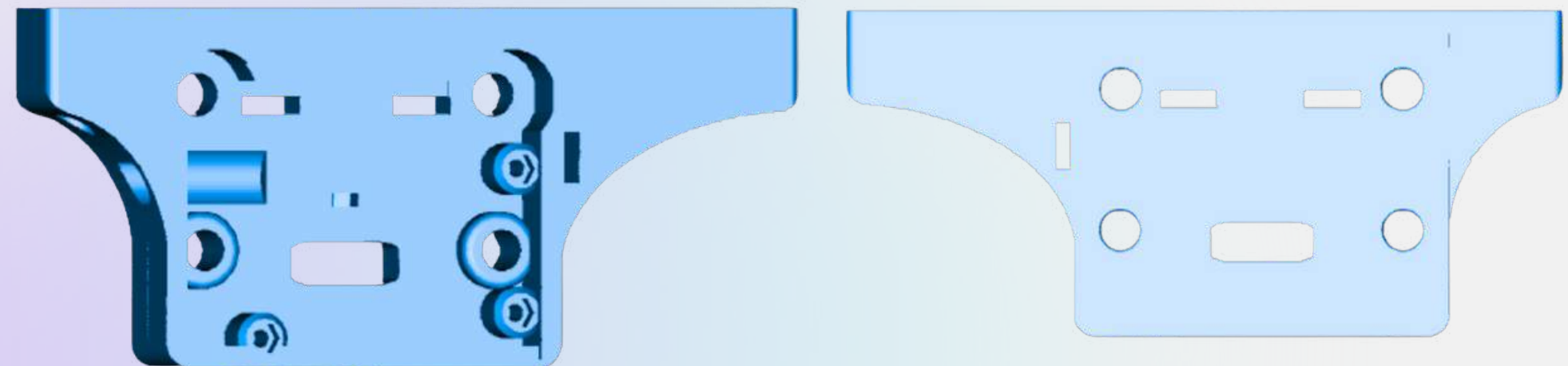
It is designed to slide along a smooth rod (using linear bearings) and support the ends of two other rods perpendicular to it.



CUSTOM COMPONENTS

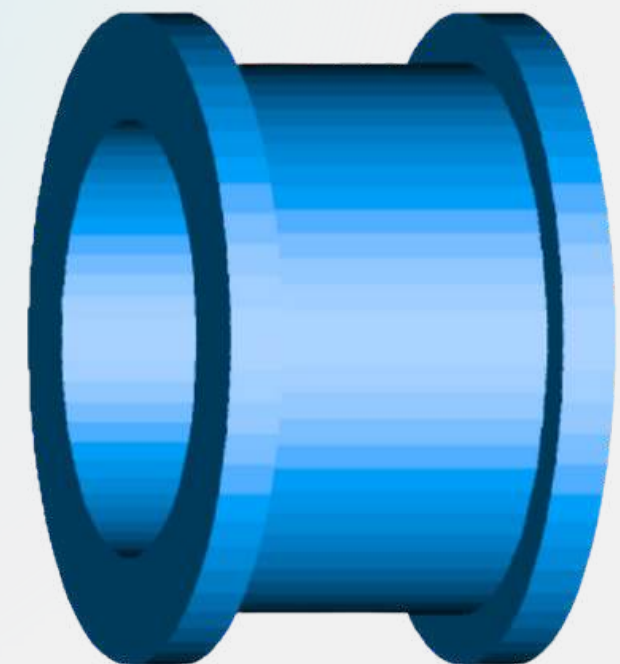
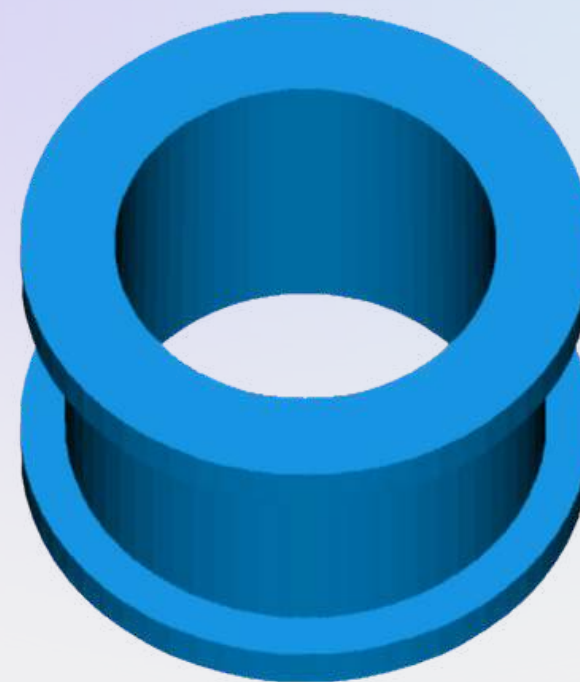
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Writing Machine Slider Y axis End.stl (The Passive Slider)

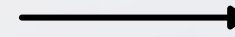


623zz Bearing Cap.stl (The Spacer)

These are Belt/Pulley Spacers. They are placed around the 623zz bearings (which act as pulleys)

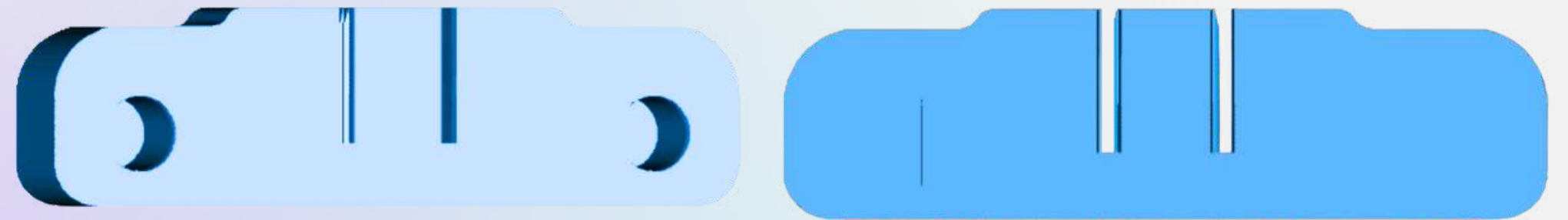


CUSTOM COMPONENTS



Writting Machine Slider X axis End.stl

Its job is to lock the smooth steel rods of the X-axis in place, preventing them from sliding out of the plastic parts during operation and keeping the frame rigid.

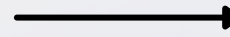


Pen Holder.stl

The Pen Holder securely grips the writing instrument—typically a ballpoint or gel pen. It is designed with a cylindrical slot and screw-tightening mechanism, allowing different pen diameters to fit firmly.



SOFTWARE INNOVATION



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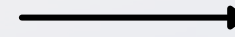
Browser-Based Control: Unlike traditional plotters requiring Inkscape + UGS + Python, our system runs entirely in Google Chrome.

Real-Time Processing:

- **Input:** "Hello World"
- **Latency:** < 500ms processing time.
- **Output:** Machine begins writing immediately.

Custom G-Code Algorithm: We wrote a JavaScript engine that generates single-stroke vector paths (Hershey Fonts) optimized for speed, rather than outlining fonts which takes double the time.

SYSTEM INTEGRATION



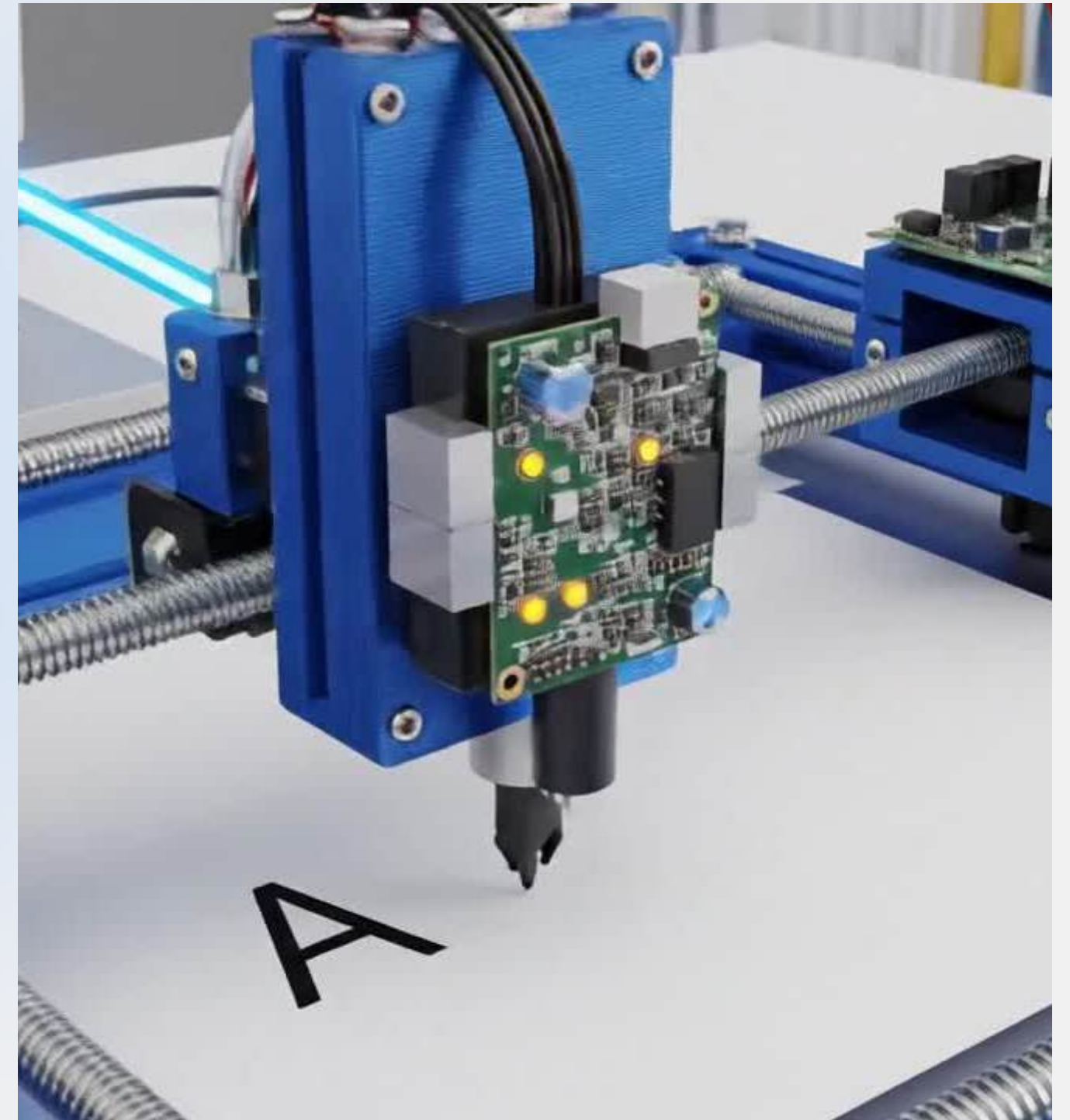
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User Speaks: "Answer A."

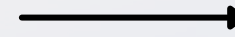
Web App: Converts audio to text "Answer A" $\rightarrow$ Looks up vector paths for 'A', 'n', 's'... $\rightarrow$ Generates G-Code G1 X10 Y10.

Browser-to-Hardware: Web Serial API opens a direct pipe to COM3 (Arduino).

Hardware Execution: Arduino receives G1, CNC Shield sends pulses to Steppers, and the pen physically writes "Answer A."



CONCLUSION



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Education (Assistive): Helping disabled students write university exams where scribes are unavailable or unaffordable.

The VocalScribe project successfully bridges the digital-physical divide, offering a solution 10x cheaper than market alternatives while adding unique Voice Control capabilities.

THANK YOU